

Course Title	The Craft of Computing		
Programme Title	B. Sc. (Honours)		
Specialization	Information Sciences		
Mode	M1	Level	1
Course ID		Credits	5
Course Type	Core	Semester	1
Version	2.0	Academic Year	2026.27
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Abstract

The Craft of Computing is the first course in the information sciences major and minor. The course is designed to be an accessible and hands-on exploration of different facets of computing. Through this exploration, we hope to initiate the transformation of students from being passive consumers of technology to active creators and problem solvers. The course features a series of activities through which students get comfortable with using a command line and become introduced to the Python programming language and the use of microcontrollers. The course will also encourage students to get to know each other and learn to work together.

Description

The Craft of Computing is a 5-credit course in the information sciences programme. It is a core course for information sciences major students in their first semester. It is also the first course in the minor sequence. The course explores different facets of computing through hands-on activities and projects.

Rationale and Introduction

The craft of computing is the first course in both the information sciences major and minor. Computing is one of the central pillars of the information sciences programme. This course is the first of several computing-related courses in the information sciences curriculum and is designed to give students an appreciation for the breadth of computing through hands-on activities and projects.

The majority of students taking this course will be in their first semester. The course is designed to cater to a diverse student body and to enable students to become accustomed to university academics. The curricular objectives of the information sciences programme include fostering technical competence along with a professional work ethic and a constructive and flexible attitude towards the use of tools. *The craft of computing* prepares students for accomplishing these objectives by laying a foundation that will be built upon in future courses and by emphasizing teamwork and self-directed learning.

The facets of computing that will be explored in this course are (i) becoming familiar with a command line interface, (ii) using a general-purpose programming language and (iii) employing microcontrollers to interface a computer with the external world. These facets have been chosen keeping in mind the hands-on nature of the course. All three facets provide ample scope for open-ended activities and projects. Moreover, there is a natural progression in complexity as one explores these facets. As a result, it is hoped that the course will not be overwhelming for a student with limited prior exposure to computing.

The information sciences programme also emphasizes a critical view towards technology and a nuanced understanding of the relationship between technology and society. These aspects are not explored in this course, but are covered extensively in *Becoming a Reflective Information Scientist* (2nd semester course).

Prerequisites

None

Indented Learning Objectives

ILO 1. Execute basic computing tasks. Students should be able to carry out computing tasks by either following specific guidelines or by identifying necessary steps by consulting manuals and other authentic sources of information

ILO 2. Foster an inclusive learning environment and responsible conduct. Students should be able to collaborate with their classmates and create a culture of sharing and care.

They should also be able to cooperate with university staff members in ensuring responsible use of facilities.

ILO 3. Examine and troubleshoot computing commands/code. Students should be able to infer and explain the output obtained after the execution of a set of commands or a computer program. They should also be able to troubleshoot failures encountered when executing computational tasks.

ILO 4. Design and implement new applications. Students should be able to apply familiar computing tasks in creative ways to generate new outcomes through a combination of tinkering and thoughtful design.

ILO 5. Introspect on their learning. Students should be able to reflect on their learning experiences and recognize when and how to seek academic help.

Syllabus and Readings

The craft of computing is designed with 3 units. Each unit is an exploration of some facet of computing and will feature several hands-on activities and a project. The first unit makes students comfortable with using a command line interface. The second unit is an introduction to the Python programming language. In the final unit, students will get some experience in working with ESP32 microcontrollers. Topics indicated as optional may be skipped in case there isn't sufficient time.

	Unit name	Weeks
Unit 1	Computing with the command line	5
Unit 2	Exploring Python	5.5
Unit 3	Interfacing with Microcontrollers	4.5

Unit 1. Computing with the command line (5 weeks)

The first unit begins with an exploration of some vignettes of computing history. Thereafter, students will learn to use a command line interface. They will use some command-line tools built for this course towards computing tasks such as creating and updating web pages with sketches. Students will also write simple shell scripts for automation tasks which will allow them to combine multiple manual steps into a single command. Through these activities, students will appreciate the utility of the Linux shell.

Class Schedule:

- Course overview, Discussion on evolution of computers (Week 1 Class)
- Preparation for group presentation, Discussion on reliable sources of technical information (Week 1 Lab session)
- Introduction to the command line (Week 2 Class)
- In-class Reflection, Group presentations (Week 2 Lab session)
- Creative use of the command line, pipes and filters (Week 3 both sessions)
- Simple shell scripts, variables, loops and environment (Week 4 both sessions)
- Accessing a remote machine with ssh and publishing web pages (Week 5 class)
- Group project (Week 5 Lab session)

Resource for group presentation

- Ignatofsky, Rachel. 2022. *The History of the Computer*. Hachette Children's Group.
- "Timeline of Computer History." <https://www.computerhistory.org/timeline/>

Reference resources

- Branch Education, 2026. The Incredible Evolution of Computers. <https://www.youtube.com/watch?v=aa6YISbAJEA>.
- "Ultimate List of Linux Bash Commands." <https://github.com/trinib/Linux-Bash-Commands>.
- "Pipes and Filters." 2005. News. Linux.Com, June 16. <https://www.linux.com/news/pipes-and-filters/>.
- "Unix Shell, Software Carpentry." <https://swcarpentry.github.io/shell-novice/>
- Shotts, William. 2026. *The Linux Command Line (Seventh Internet Edition)*. Available at <https://linuxcommand.org/tlcl.php>

Unit 2. Exploring Python (5.5 weeks)

In this unit, students will progress from the command line to a general purpose programming environment. Students will get introduced to the Python programming language and will learn the basics of structured programming. As students progress through the unit, they would be able to build more complex tools than the first unit. They would also be able to appreciate what goes on "behind the scenes" when these tools are used.

Class Schedule:

- Introduction to Python, basic expressions, datatypes and writing scripts (Week 6 both sessions)
- Iterations & Conditionals (Week 7 both sessions)
- Functions & Modules (Week 8 both sessions)

- Working with Files & APIs (Week 9 both sessions)
- Group project (Week 10 both sessions)
- In-class test (Week 11 class)

Resource for projects

- Sweigart, Al. 2021. *The Big Book of Small Python Projects*. No Starch Press

Reference resources

- Chapters 1-11 from Downey, Allen B. 2024. *Think Python, 3rd Edition*. O'Reilly Media
- David Beazley & Brain K. Jones 2013. *Python Cookbook, 3rd Edition*. O'Reilly Media
- Chapter 0-3 from David Beazley. *Practical Python Programming*. Available at <https://dabeaz-course.github.io/practical-python/Notes/Contents.html>

Unit 3. Interfacing with Microcontrollers (5 weeks)

In the final unit of the course, students will explore how to interface their computer to the external world using a microcontroller and different sensors and output devices. Students will use ESP32 microcontrollers and program them using microPython. As students work on different activities in this unit, they will recognize that computation need not always be abstract.

Class Schedule:

- Introduction to ESP32 and the Thonny Integrated Development Environment (Week 11 Lab session)
- Basics of electronic circuits (Week 12 both sessions)
- Integrating sensors and output components (Week 13 both sessions)
- In-class Reflection (Week 14 class)
- Group project (Week 14 Lab session and both sessions in assessment week)

Reference resource

- Monk, Simon. 2024. *Programming ESP32: Learn MicroPython Coding and Electronics (Second Edition)*. Monkmakes Press.

Pedagogy

This course will largely be transacted through in-class activities and project-based learning. Working in groups will be an essential part of the course. Student diversity will be addressed through group work and academic support provided by peer tutors or associates. Reasonable support will be made available for students who have applied for formal academic accommodations as per institutional norms.

Assessment

The assessment methodology for the course is summarized in the table below:

Assessment Type	Number of assessments in the semester	Weightage of each assessment	Contribution to overall score	Relevant ILOs
In-class Reflections	2	4% + 7%	11%	5
Group Presentation	1	5%	5%	2
Group Projects	3	5% + 10% + 10%	25%	1, 3, 4
Activity Reports	Best 3	3%	9%	1, 2
Test	1	20%	20%	3
Learning Journal	1	20%	20%	1, 3, 5
Class Participation	1	10%	10%	2, 5

- **In-class Reflections:** Students will be asked to write reflections at the beginning (week 2) and end (week 14 or assessment week) of the course. The first reflection will ask them to write about their views on computing and what they would like to learn about computing. The second reflection will ask them to look back at the course and revisit their views on computing. They would also be asked to introspect on the challenges they encountered and how they addressed them.
- **Group Presentation:** The class will be divided into groups (of ~4 students) and each group will explore some vignette of computing history using the illustrated book *The History of the Computer* by Rachel Ignatofsky as a starting point. Apart from enabling sharing of information, the presentation will also allow students to get to know each other. The presentation will be scheduled early in the semester (week 2 or 3).
- **Group Projects:** In each unit, students will work in groups (of 3 or 4) on an open-ended project based on the computing facets introduced in that unit. Class time will be dedicated for students to work on the project and to seek help from the instructor.

- **Activity Reports:** Students will be carrying out several hands-on activities in pairs or groups of 3 during lab sessions. In each session, one group member will be responsible for ensuring that they complete all assigned tasks. This member would have to submit a brief report about the progress the group made and whether any of the group members faced difficulties in completing the tasks. Over the semester, each student will submit 3 or 4 reports and the best three will be counted.
- **Test:** An in-class test will be conducted after unit 2 has been completed (week 11 or 12). This will be based on the first two units and would assess the ability of students to predict outputs of commands/Python code and to identify modifications to be made to obtain a particular output.
- **Learning Journal:** Students will be asked to maintain an online journal documenting their learning experiences over the course. The journal will be created from Markdown files using a static site generator. While students would be free to write as many entries as they wish to, the following minimal set of six would be required: (i) an entry related to their group presentation, (ii) three entries documenting their experiences in the three group projects and (iii) two entries related to any troubleshooting they had to carry out during class activities.
- **Class Participation:** Class participation would be assessed based on efforts to facilitate collaborative learning, responsible use of facilities and consistent active engagement with the course. Students who proactively seek academic support will be considered actively engaged.